Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Partha Mukherjee	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	585 Purdue Mall, West Lafayette, IN 47907-2088			

2. Technology Overview

Title of Proposed Technology	**Next-generation AI BMS platform based on Sensor fusion**			
Korea's 12 major national strategic technology fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (Major Category)**		**Technology Field 2 (Major Category)**	
	Electrical / Electronics		***Information & Communication***	
	Medium category	Small category	Medium category	Small category
	Batteries	***Secondary Batteries***	***Software***	***S/W Solutions (Software Solutions)***
Proposed technology concept	**Core Concept** The proposed technology develops a "Next-Generation AI Battery Management System (BMS) with Sensor Fusion" that combines Sensor Fusion and Physics-Informed Machine Learning (PIML) to detect thermal runaway and internal anomalies that may occur in batteries of Energy Storage System (ESS) and Electric Vehicle (EV) at an early stage. The core concept is to operate and verify it in a cloud-edge collaborative structure. In particular, the developed diagnostic/alarm performance is reproducibly verified based on the Purdue's Center for Advances in Resilient Energy Storage (CARES) verification system (test protocol, scenario, and evaluation index) to ensure reliability. ● Data Flow (Sensor Fusion-Based) Collects basic BMS data (voltage, current, temperature, etc.) from field and EV batteries, as well as additional sensor data (e.g., multi-point temperature, insulation /			

	impedance, EM, etc.) as needed. The collected raw data undergoes time synchronization, noise removal, outlier filtering, and feature extraction at the edge, and is then transmitted to the cloud in the form of summarized data/event logs. This reduces communication bandwidth burden while preserving precursor signals. ● Hybrid Architecture (Edge Early Warning + Cloud Enhancement) The edge resides on-site and handles low-latency early warning (predictive score/abnormality flag) and event buffer recording (pre- and post-accident data). While the existing BMS retains the final authority for protection and control (output limitation/blocking, etc.), the system is designed to leverage the edge's predictive indicators for faster and more sophisticated safety responses. The cloud performs sensor fusion/PIML model learning/validation/ version management based on data collected from various sites/vehicles and deploys validated models and policies to the edge to continuously enhance system performance. This architecture is a strategy that simultaneously secures on-site safety (independent operation/low latency) and enhanced central analysis (large-scale data learning/validation). **Key Functions and Potential Applications** ● Core features (focused on thermal runaway precursor detection) 1. Thermal runaway precursor detection: Provides precursor scores, risk stage classification, and early warning (lead time). 2. Abnormal Cause Estimation/Explanation Diagnosis: Interpreting abnormal patterns in a physically consistent manner using Physical Constraint-Based Matrix. 3. False alarm rate management and field adaptation: Threshold/model tuning based on operational data, field deviation compensation (cell / module / pack, temperature / aging). 4. Event-based black box recording: Automatically stores data before and after an abnormal event for post-event analysis and model improvement. ● Verification and Reliability Ensuring (CARES-based) The developed diagnostic and alarm performance is repeatedly verified using test scenarios (operating conditions/defect situations) and evaluation indicators (lead time, false alarm rate, recall, etc.) that comply with the CARES verification system, thereby ensuring reproducible safety performance rather than a simple demonstration. ● Scalability (Fleet/Multi-site Application) Designed with standard APIs and a modular structure, it can be applied to various battery systems such as ESS and EV, and ensures scalability to enable fleet-level performance monitoring and model updates (OTA) for multiple sites/vehicles.
Rationale for Joint R&D Project	**Critical Technical Barriers** ● Technical necessity Currently, many BMSs are optimized for local control and protection logic centered on voltage, current, and temperature. This limits early detection (ensuring lead time) and

reproducible verification of rare and high-risk events such as thermal runaway and internal anomalies. Furthermore, as field variations (cell/module/pack configuration, operating temperature, and aging status) increase, models based on a single data source become less generalizable, which can lead to increased false alarm rates or false positives. Therefore, it is essential to combine multi-sensor-based sensor fusion with physics-informed machine learning (PIML), which incorporates the physical, thermal, and electrochemical constraints of batteries, to implement an "explainable and reliable" anomaly detection model. Furthermore, it is essential to secure technology that simultaneously achieves low-latency on-site alarms and centralized learning/verification through a cloud-edge collaborative architecture.

- Industrial necessity

  The BMS market is experiencing rapid growth due to the proliferation of EVs and ESS, and battery safety accidents (fires/thermal runaway) have emerged as a key risk for companies, causing massive financial losses, operational interruptions, insurance and regulatory risks, and damage to brand trust. Accordingly, the industry is demanding highly reliable diagnostics (AI+BMS) that go beyond simple protection functions and enable early warning of thermal runaway warning signs and traceability. This technology can be expanded into an operational BMS that detects thermal runaway warning signs early, substantially reducing the probability of accidents and the extent of damage. It also manages false alarm rates based on field operation data while maintaining detection sensitivity, creating significant economic value in terms of safety, operating costs, and uptime.

- Policy/Social Necessity

  As the "Carbon Neutral 2050" goal, renewable energy expansion, and electric vehicle adoption accelerate, batteries will become a core part of the national energy and mobility infrastructure, making battery safety a prerequisite for a social safety net. In particular, as the scale of ESS installations increases, the potential for damage in the event of an accident increases, necessitating safety technologies focused on early warning and prevention. This technology contributes to resource circulation and the sustainable energy transition by preventing and reducing thermal runaway accidents, lowering social risk, and enhancing the reliability of battery condition diagnosis, thereby providing a basis for reuse and recycling decisions (life prediction/residual value assessment).

Why Global Collaboration?

While Korean SMEs possess cloud-edge service implementation, data pipeline, field application, and MLOps operation capabilities, it needs the core technology to observe, interpret, and reproducibly verify the internal state of a battery, which is key to detecting precursors of thermal runaway. Purdue University, on the other hand, possesses an irreplaceable capability to “observe and verify precursor signals” based on electromagnetic (EM)-based precision sensing and CARES (linked to UL Research Institutes), a battery safety science and testing and verification ecosystem. Without Purdue’s EM precision sensing and CARES-based verification system, the collected data would remain limited to

(max 10 pages)

	observations such as voltage and surface temperature, making it difficult to sufficiently secure and verify the early detection performance (lead time, false alarm rate, and false detection rate) of critical defects such as precursors of internal short circuits and localized heating. Therefore, when Korean SME's operational platform, edge real-time alert implementation, and MLOps capabilities are combined with global research institute's source sensing, PIML advancement, and CARES validation protocol, it is possible to implement innovative technology that can be commercialized as a next-generation BMS with 'safety performance proven' that goes beyond the laboratory level and is demanded by the global market.
Key words	1. Electric vehicles (EVs) and energy storage systems (ESS) 2. Early Warning of Thermal Runaway 3. Sensor fusion and electromagnetic (EM) precision sensing 4. Diagnosis/prediction based on PIML (Physics-Informed Machine Learning) 5. Cloud-edge collaborative BMS based on the CARES verification system

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

R&D Objectives

To develop a sensor fusion and PIML system for thermal runaway precursor detection based on the CARES validation system. The goal is to develop a next- generation, high-precision, high-reliability BMS that secures early warning lead time and reliability (reduction of false alarms and misdetections) in various battery operation environments such as EVs and ESS by implementing "technology" and "cloud-edge collaborative BMS operation and deployment architecture."

Detailed Objectives

- Development of a next generation BMS capable of detecting thermal runaway precursors

  We are implementing an intelligent BMS capable of early detection of thermal runaway and internal anomalies that may occur in EV and ESS batteries, and enabling on-site response (alarm/driving restriction recommendation/inspection trigger) by grading the risk level.
- Sensor fusion-based anomaly diagnosis

  In addition to basic BMS data such as voltage, current, and temperature, if necessary, extended sensors (multi-point temperature, insulation/impedance, EM, etc.) are fused to strengthen precursor signals and enable stable diagnosis even under various operating conditions and aging states.
- Securing a highly reliable diagnosis and prediction algorithm based on PIML

  By applying PIML (Physics-Informed ML) that reflects the physical, thermal, and electrochemical constraints of batteries, we secure explainable and generalizable SOC/SOH estimation and anomaly detection performance even in rare failure situations.
- Building a cloud-edge collaborative operating system

  The edge handles low-latency early warning and event buffer recording (pre- and post-accident logs), while the cloud performs large-scale data-based learning / verification / versioning / policy distribution (OTA) to maintain operational reliability even in the face of changes in communication quality.
- CARES-based verification and performance validation

  Developed precursor detection and diagnostic performance is repeatedly verified using test scenarios/protocols and evaluation indicators compliant with the CARES verification system, ensuring reproducible safety performance rather than a simple demonstration.
- Build an operator portal and alert system

  Implement a dashboard and alert/report system that allows operators to intuitively view battery status, precursor scores, risk levels, alert history, and maintenance recommendations.

(max 10 pages)

R&D Target Specifications

The existing "fault and anomaly detection accuracy (≥90%)" is vulnerable to attack during the evaluation, so it includes KPIs (lead time, false alarm rate, recall) that are suitable for the early warning task, and the numbers (X/Y/Z) can be finalized during the task.

Key Performance Indicators	Unit	Final goal	Evaluation method
SOC estimation error	%	MAE within ±2%	Certification Authority Evaluation
SOH estimation error	%	Within ±5%	Certification Authority Evaluation
Charge/discharge efficiency (system standard)	%	≥ 95%	Certification Authority Evaluation
Lead time for thermal runaway precursor detection	minutes/seconds	≥ X minutes (or X seconds) advance warning	CARES-based test scenarios + certified assessment
Recall of detection of major anomalies (precursors)	%	≥ Z%	CARES-based test scenarios + certified assessment
False Alarm Rate (FPR)	%/day or cases/day	≤ Y%/day (or ≤ Y cases/day)	Long-term operational data verification + certified evaluation

R&D Plan

The step-by-step research and development plan to achieve the goal is as follows.

Step	Period (months)	Development details	Performing entity
1	1–3	Define requirements and safety KPIs, analyze the target (ESS/EV) operating environment, define data schema and events, and align the CARES verification scope.	Led by the host organization, cooperation between joint organizations
2	4–6	Sensor configuration and edge collection design (BMS + extended sensors), time Synchronization, filtering, feature design, edge-BMS interface definition	Cooperation between host/joint organizations
3	7–9	Draft Cloud-Edge Collaboration Design, Data Pipeline and Model Deployment Policy, and CARES Test Scenarios	Cooperation between host/joint organizations
4	10–13	Sensor fusion and PIML model development/training/verification (phase 1), SOC/SOH/precursor detection performance evaluation	Jointly led by a joint organization, supported by a main organization
5	14–16	Edge real-time precursor detection implementation, lightweight model, event buffer recording, and cloud integration.	Cooperation between host/joint organizations
6	17–19	Establishing advanced ModelOps (MLOps), automated deployment and rollback, and performance monitoring	Led by the host organization, cooperation between joint organizations
7	20–22	Establishment of an operational portal and alert system, external RESTful API integration	Host organization (Korean SME)
8	23–27	Verification (primary) integration testing, long-term operational verification, model and threshold tuning	Cooperation between host/joint organizations
9	28–30	Model and sensor fusion enhancements (2nd phase), improved pre-processing performance, and edge optimization	Jointly led by a joint organization, supported by a main organization

(max 10 pages)

	10	31–33	CARES-based performance verification confirmed, KPI final achievement verification	Joint organization leadership, cooperation between leading organizations
	11	34–35	Responding to public certification, improving product quality, and organizing operational guides	Led by the host organization, cooperation between joint organizations
	12	36	Organizing technical data, reporting final results, and preparing business-related materials	Cooperation between host/joint organizations

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Purdue University's Roles and Responsibilities (R&R): Building a Technological Moat ● Development of core technologies (sensing, modeling, and lightweighting) ■ Development of EM-based non-contact precision sensing technology ■ Establishment of an EM sensing/measurement methodology sensitive to internal battery abnormalities (current path changes, local heating, etc.) ■ Guidance on sensor configuration, measurement conditions, and signal processing (preprocessing/feature extraction) that can be combined with BMS basic data (voltage, current, temperature, etc.) and extended sensors (multi-point temperature, insulation/impedance, etc.) ■ Design of a sensor fusion-based thermal runaway precursor detection model ■ Design of a sensor fusion architecture (feature-level/decision-level) and risk scoring system that fuses heterogeneous data from multiple sensors ■ Establishment of alert tiering/threshold design principles that optimize early warning performance in terms of lead time, false alarm rate, and non-detection rate ■ Development of a high-precision condition estimation and deterioration model based on PIML ■ SoX if necessary and development of deterioration/abnormality models using Physics-Informed ML (PIML) that reflects physical, thermal, and electrochemical constraints ■ A hybrid model (physical model + ML) and regulation term (loss) design is proposed to maintain generalization performance even in situations of data shortage/field variation (temperature, aging, pack structure) ■ Providing lightweight models and inference optimizations for edge deployment ■ Lightweight (model compression/latency optimization) and inference pipes to enable low-latency operations in edge environments. Line design ■ Providing algorithm packages/reference implementations in a deployable form that d efines the outputs (predictive scores/risk levels/descriptive indicators) required at the edge ● Verification and reliability assurance (CARES-based verification system) ■ CARES-based verification protocol design ■ Design of test scenarios, condition matrices, and data collection specifications for

	verifying thermal runaway/internal short-circuit precursor detection performance based on the CARES (ULRI-linked) ecosystem ■ Presentation of test procedures (repeatability/reproducibility), quality criteria, and data management principles for reproducible validation beyond the laboratory level ■ Defining standard KPIs and establishing an objective evaluation system ■ Defining key KPIs (lead time, false alarm rate (FPR), critical defect recall, and long-term operational stability) appropriate for early warning tasks, rather than simply focusing on "accuracy" ■ Provides KPI calculation methods (based on window/event definition/labeling) and performance reporting formats ● Global linkage support (North American expansion and expansion of results) ■ North American verification/certification requirements advisory ■ Technical advice and testing/certification roadmap proposals for the safety, verification, and regulatory environments required for entry into the North American market ■ Technical documentation (English) framework support for global customer/agency communication ■ Creating international joint achievements ■ Contribution and specification of core technologies (sensing/model/verification protocols) for securing international joint patents ■ Support for objective recognition and dissemination of project results through joint pa per publication and international academic/industry network connections
R&R of Korean SME partners	Korean SME's Role and Responsibilities (R&R): Accelerating commercialization and implementing go-to-market (minimum security) initiatives. ● Platform development and integration (implementation of operational products) ■ Building a Cloud-Edge Collaborative BMS Operation Platform · Design and Implementation of a Cloud-Edge Hybrid Architecture for a Next-Generation BMS with Thermal Runaway Predictor Detection · Implementation and stabilization of low-latency precursor score calculation, alerting, and event buffer (pre- and post-accident log) recording functions at the edge · Establishing an existing BMS linkage (CAN/interface): Establishing an edge (recommendation/flag) + BMS (final protection action) structure ■ Establishment of Data pipeline and MLOps (ModelOps) operating system · A multi-sensor/BMS data collection/purification/synchronization/quality control and long-term storage pipeline · ModelOps/MLOps implementation including model versioning, validation gates, OTA deployment/rollback , and operational monitoring (drift /false alarm rate, etc.) ■ Development of an integrated operation portal (UI/UX) and alert system · Implemented dashboard/alarm/reporting features that provide precursor scores, risk levels, alert history, event reports, and inspection recommendations

(max 10 pages)

	・ Providing a standard API (to the extent necessary) for linking with external systems ■ Security (only the minimum required items are reflected) ・ Apply essential security requirements such as terminal authentication, transport layer encryption (TLS), OTA integrity (signature), and rollback level ● Verification, productization, and commercialization (field verification and market application) ■ Establishment and operation of a domestic verification test bed ・ ESS/EV (or simulated environment) demonstration operation and data collection ・ Model/threshold tuning and field deviation correction to achieve KPIs (lead time, false alarm rate, recall) ■ Establishing product commercialization and customer application scenarios ・ Establishing installation, operation, and maintenance processes and establishing a roadmap for transitioning from PoC to commercial use ・ Development of commercialization strategies and sales/marketing materials (focused on effectiveness and ROI) ● Overall project management (role of the lead organization) ■ Schedule/budget/performance management, risk management ■ Reporting, administration, performance indicator management, and final output organ ization in accordance with TIPA requirements

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Expected outcomes ■ Securing core technology for thermal runaway warning detection: Securing next-generation BMS core algorithms that detect internal abnormalities in EV/ESS batteries early and calculate risk levels based on sensor fusion and PIML. ■ Establishing an edge early warning + cloud-based advanced architecture: Securing an operational architecture that performs low-latency precursor scoring, alerts, and event buffers (pre- and post-accident logs) at the edge, and enables learning/verification/version management/distribution (OTA/rollback) in the cloud. ■ Establishing a CARES-based verification framework: Define standard test scenarios and evaluation indicators (lead time, false alarm rate, recall, etc.) for key safety functions such as thermal runaway/internal short-circuit warning, and establish a system that enables repeatable verification. ■ Ensuring application scalability: Building a model/feature library and data schema applicable to various BMS/battery configurations (cells, modules, packs) and operating conditions (temperature, aging). ■ Intellectual Property (IP) and Technology Barriers: Creating domestic and international patents and academic papers focusing on key elements such as sensor fusion, PIML, edge-mounted optimization, and verification protocols. ● Economic and industrial impact

	■ Reduce accident risk and loss costs: Reduce the likelihood of safety accidents and the extent of damage through early warning (ensuring lead time) and false alarm rate management, and reduce recall, downtime, insurance/regulatory risks. ■ Improving asset life and TCO: Precision diagnostics-based operational optimization improves battery life and availability, reducing total cost of ownership (TCO) for EV fleet/ESS operators. ■ O&M Efficiency: Event-based maintenance (condition-based maintenance) and remote diagnostics improve maintenance efficiency and reduce operating costs. ■ Strengthening market competitiveness: By securing safety-focused BMS technology with proven performance, we simultaneously secure technological reliability and barriers to adoption in the global BMS/ESS market. ● Social and environmental impact ■ Strengthening the social safety net: Reduce casualties and property damage and improve social safety by preventing battery fires and thermal runaway accidents. ■ Enhancing the stability of the energy transition: Improving ESS Safety Contributes to Grid Stabilization as Renewable Energy Expands, strengthening the foundation for carbon neutrality ■ Promoting resource circulation: By improving the reliability of battery condition (SOH, etc.) diagnosis, the accuracy of reuse/recycling decisions is increased, thereby activating the resource circulation market. ■ Reduced environmental impact: Reduces waste battery generation and contributes to reducing waste and carbon emissions by extending lifespan and optimizing replacement cycles.

6. References

(1) Rose, S., Borchert, O., et al. (2020). *Zero Trust Architecture (NIST SP 800-207)*. National Institute of Standards and Technology.

(2) UL Research Institutes (ESRI). *Center for Advances in Resilient Energy Storage (CARES)*. Retrieved from https://ul.org/institutes-offices/electrochemical-safety/center-for-advances-in-resilient-energy-storage-cares

(3) Purdue University Mechanical Engineering News (2023). *Purdue joins forces with UL Research Institutes to create new battery safety research center (CARES)*. Retrieved from https://engineering.purdue.edu/ME/News/2023/purdue-joins-forces-with-ul-to-create-battery-safety-research-center

(4) Grand View Research. *Battery Management System (BMS) Market Size, Share & Trends Analysis Report*. Retrieved from https://www.grandviewresearch.com/industry-analysis/battery-management-system-bms-market

(5) International Society of Automation (ISA). *ISA/IEC 62443 Series of Standards overview*. Retrieved from https://www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series-of-standards

(6) Circularize. *Battery Regulation EU: Learn about battery passports*. Retrieved from https://www.circularise.com/blogs/battery-regulation-eu-what-you-need-to-know-about-battery-passports

(7) Purdue College of Engineering. *Diagnosing the true temperature of lithium-ion batteries*. Retrieved from https://engineering.purdue.edu/ME/News/2025/diagnosing-the-true-temperature-of-lithiumion-batteries